

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 1 of 6

(1) Research Program/Project/Study Title: Density and Distribution of <i>Mytella strigata</i>: A Comprehensive Study in Pilar, Capiz	
Proponent Name:	Ianthe Marie P. Benliro
Academic Rank:	Asso. Prof. III
Co-Authors (with academic rank):	Gelene Roda I. Alvarez (Instructor I), Dr. Rey J. dela Calzada (Prof. IV), Robert Christian F. Patani (Instructor I) and Alessandra Nicole B. Falalimpa (Instructor I)
Campus/Satellite College/Address:	Pontevedra
Telephone/Cell Phone/Fax Number:	09338769116
E-mail Address:	ibenliro@yahoo.com

Program or Project has component (if no proceed to 2)? ☐ / Yes ☐ No

Component Title:	Managemen of <i>Mytella strigata</i> in Pilar, Capiz
Component Leader:	Gelene Roda I. Alvarez
Academic Rank:	Instructor I

Component Title:	Development of Innovative Products from <i>Mytella strigata</i>
Component Leader:	Alessandra Nicole B. Falalimpa
	Instructor I

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ / Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ /FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation	
Municipality/City:	Pilar
Province:	Capiz
Region:	VI

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 2 of 6

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Aquatic Invertebrates <i>Mytella strigata</i> (Black Mussel)		(8) Discipline/Field of Specialization Fisheries/biodiversity/invasive species
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) This study would serve as baseline and provide the first record of the invasive <i>Mytella strigata</i> in the coastal area of Pilar, Capiz. The Municipality of Pilar, Capiz is one of the important fishing grounds in the Province of Capiz. Local fishers derive food and income not only offshore but also from the vast intertidal zone during lowtide. In the previous years, population of <i>Mytella strigata</i> (black mussel) which is considered as a highly invasive species (Joyce et al., 2023; Fabiosa et al., 2021; Yu et al., 2023; Biju Kumar et al., 2019; Huang et al., 2021) was not evident in Bang-ogay area. Recently it has been noticed that these organisms are already present in dense mats covering the soft sandy muddy substrate. The black mussel population was observed beneath the natural mangrove patches consisting of <i>Sonneratia alba</i> and <i>Avicennia marina</i> along the area where Balisong creek was forced to detour when the natural outflow has been blocked by sediments that accumulated as a result of human intervention in the nearby resort. The study area is particularly of interest because it was subjected to abrupt physical changes in terms of alteration of current direction. Unfortunately there was no baseline data on physico-chemical parameters and plankton dispersal in the area. Since black mussels were not very common in the area in the past it may be possible that the change in water circulation may have influenced its proliferation at present. It was stipulated that many invasive mollusks may have been brought by ballast water from ships (Spinuzzi et al., 2013) but other studies indicated that their presence were first observed in aquaculture ponds (Lodeiros et al., 2021). Some assessments have been done in Manila Bay and Mindanao (Fabiosa et al., 2021; Fuertes et al., 2021; Mendoza et al., 2023), but there is no published research that documented the <i>Mytella strigata</i> in Pilar, Capiz, hence this study.		
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.) The research team will conduct an ecological assessment of the area where the <i>Mytella strigata</i> (black mussels) abound in Pilar, Capiz. A location map showing the study site was downloaded from Google Maps (2023 and 2021). It clearly shows the amount of sand that has accumulated across the exit of water effectively diverting the creek. This occurrence together with other factors may be examined in relation to the recent proliferation of black mussels in the study site. The transect-quadrat method will be used to determine the density of black mussels. A caliper and digital weighing scale will be used to measure the size structure and weight of the mussels. A refractometer, thermometer and pH meter will be used to measure temperature, salinity and pH of the water. A digital current meter will be used to measure water flow and current direction. Likewise soil samples will be gathered for laboratory analysis.		
(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.) This study aims to assess the <i>Mytella strigata</i> (black mussel) in Pilar, Capiz. Specifically it aims to: - determine the abundance in terms of density and distribution of the black mussels in Pilar, Capiz; - determine the size structure of the black mussels in Pilar, Capiz. - verify the physico-chemical parameters including the substrate characteristics in the study site.		

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 3 of 6

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology/ information from which the project proposal will take-off).

The sudden overabundance of invasive species such as *Mytella strigata* has alarmed conservationists because it has already invaded many countries such as Philippines (Fabiosa et al., 2021; Fuertes et al., 2021; Mendoza et al., 2023), Hongkong (Joyce et al., 2023), Singapore (Lim et al., 2018), China and Indo-west Pacific (Yu et al., 2023; Ma et al., 2022), Taiwan (Huang et al., 2021), Southeast Asia (Sanpanich and Wells, 2019), and Indian waters (Biju Kumar et al., 2019; Jayachandran et al., 2019). Invasive species are known to displace native species and may also affect the livelihood of locals. On the other hand it can be a source of raw materials for developmental studies considering the potential of bioactive compounds derived from black mussels. There is great interest in mussels lately because of the bioactive peptides found in mussels that may have tremendous pharmaceutical and other commercial/industrial applications (Jeyasanta et al., 2020).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as variables, treatments, experimental procedures and design, statistical analysis, evaluation methods, observations, and strategies for implementation.

This is a descriptive study which will be conducted at geographical location 11° 40' 49" N, 123° 19' 29" E. The abundance will be determined using transect-quadrat method covering the area where the mussels can be observed seaward from the shore. Periodic measurement of mussel sizes from the smallest to the largest will be done quarterly within a year. The size structure in a clump will likewise be noted. Physico-chemical parameters will be recorded during each sampling event. Current speed and direction will be determined using digital current meter. Inundation will likewise be noted. It will be a combination of quantitative and qualitative study which will include secondary data through informal interviews with the gleaners frequenting the area regarding their observations and the impact of the invasive species on their livelihood.

(14) Major Activities/Workplan

Activities/Research Objectives	Q1	Q2	Q3	Q4
a) determine the abundance in terms of density and distribution				
b) determine the size structure of the black mussels in a mudflat in Pilar, Capiz.				
c) verify the physico-chemical parameters in the study site.				
Data Analysis				
Report Writing				

(15) Expected Output

The study will serve as baseline for future ecological studies in the estuary of Pilar, Capiz. At the same time, the location and population density of wild stocks may be established as a requirement for resource management and recommendations to the LGU for possible crafting of policy.

Utilization of the black mussel and product development may also be done to provide alternative livelihood to the community.

(16) Technology Transfer Activity Design

- Recommendations for Management Policy
- Development of innovative products utilizing the shell or the meat of the invasive organism.

(17) Target Beneficiaries

LGU, local fishers, conservationist and researchers

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 4 of 6

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Ianthe Marie P. Benliron	Study Leader	Marine Fisheries
Gelene Roda I. Alvarez	Component 1 Leader	Aquaculture
Rey J. dela Calzada	Co-researcher	Aquaculture
Robert Christian F. Patani	Co-researcher	Aquaculture
Alessandra Nicole B. Falalimpa	Component 2 Leader	Post Harvest Fisheries

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	2 field assistants 500 x 2pax X 4qtr	PhP 4,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance	50 + 20=70x2 x5pax x4qtr	PhP 2,800.00
- Meal allowance during data gathering	250 x 5pax x 4qtr	PhP 5,000.00
Soil Analysis		PhP 1,000.00
- Miscellaneous (for incidental expenses)		PhP 3,600.00
- Printing	2 reams Bond 20 / 70 Gsm A4 Bond Paper @ 200 Ink EPSON Printer 3 sets @ PhP 1,500.00	PhP 400.00 PhP 4,500.00
C. CAPITAL OUTLAY		
Digital Multi-parameter analyzer	1 unit@ PhP 45,000.00	PhP 45,000.00
Powerwinder waterproof fiberglass measuring Long Tape (100m)	1 unit@ PhP 1700.00	PhP 1,700.00
Vernier Caliper	2 units @ PhP 1,500.00	PhP 3,000.00
Digital Milligram Pocket Scales 0.001g X 50g.	1 unit @PhP 3,500.00	PhP 3,500.00
Digital Current Meter	1 unit @PhP 15,000	PhP 15,000.00
Folding Writing Slate (waterproof)	5units@PhP400	PhP 2,000.00
Total		PhP 91,500.00

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 5 of 6

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Literature Cited

- Biju Kumar, A., Ravinesh, R., Oliver, P.G., Tan, S.K. and Sadasivan, K. 2019. Rapid Bioinvasion of alien Mussel *Mytella strigata* (Hanley 1843) (BIVALVIA: MYTILIDAE) Along Kerala Coast, India: Will this Impact the Livelihood of Fishers in Ashtamudi Lake? Journal of Aquatic Biology & Fisheries | Vol. 7 | 2019 | pp. 31 – 45
- Blakemore, K. 2016. DOCCM-2595952 Marine: quadrats for invertebrate and macroalgal communities v1.0 Inventory and monitoring toolbox. Department of Conservation. New Zealand Government. 34p. <https://www.doc.govt.nz/globalassets/documents/science-and-technical/inventory-monitoring/im-toolbox-marine-quadrats-for-invertebrate-and-macroalgal-communities.pdf>
- Carlton, J.T., 1996. Pattern, Process and Prediction in Marine Invasion Ecology. Biological Conservation 78 (1996) 97-106 Copyright © 1996 Elsevier Science Limited Printed in Great Britain. All rights reserved 0006-3207/96
- de Souza, T.O., et al., Population structure and identification of two matrilinear and one patrilinear mitochondrial lineages in the mussel *Mytella charruana*, Estuarine, Coastal and Shelf Science (2014), <http://dx.doi.org/10.1016/j.ecss.2014.11.009>.
- Fabiosa, M.P., Abao, C.A.A., Fabiosa, N.P., and K.I.N. Uba. 2021. First Record of the Invasive Mussel *Mytella strigata* (Mollusca: Bivalvia: Mytilidae) in Panglil Bay, Southern Philippines. Journal of Environment and Aquatic Resources. Vol. 6, Pages 35-46. <https://doi.org/10.48031/msunjeat.2021.06.03>.
- Fuertes, V.N.B., de los Reyes, R.A. and H. M. Monteclaro. 2021. The spread of the non-indigenous mussel species *Mytella strigata* (Hanley, 1843) in the Philippines: Ensuing issues and responses of local communities. Regional Studies in Marine Science 41 (2021) 101576.
- Huang YC, Li ZK, Chen WL, Chan CC, Hsu HY, Lin YT, Huang YS, Han YS (2021) First record of the invasive biofouling mussel *Mytella strigata* (Hanley, 1843) (Bivalvia: Mytilidae) from clam ponds in Taiwan. BioInvasions Records 10(2): 304–312, <https://doi.org/10.3391/bir.2021.10.2.08>
- Jayachandran PR, Aneesh BP, Oliver PG, Philomina J, Jima M, Harikrishnan K, Bijoy Nandan S (2019) First record of the alien invasive biofouling mussel *Mytella strigata* (Hanley, 1843) (Mollusca: Mytilidae) from Indian waters. BioInvasions Records 8(4): 828–837, <https://doi.org/10.3391/bir.2019.8.4.11>
- Joyce PWS, Lee SY, Falkenberg LJ (2023) First record of the alien invasive mussel *Mytella strigata* (Hanley, 1843) in Hong Kong. BioInvasions Records 12(2):385–391, <https://doi.org/10.3391/bir.2023.12.2.03>
- Lim, J.Y., Tay, T.S., Lim, C.S., Lee, S.S.C., Teo, S. L.M. and K.S. Tan (2018): *Mytella strigata* (Bivalvia: Mytilidae): an alien mussel recently introduced to Singapore and spreading rapidly, Molluscan Research, DOI: 10.1080/13235818.2018.1423858 To link to this article: <https://doi.org/10.1080/13235818.2018.1423858>
- Lodeiros, C., Hernández-Reyes, D., Salazar, J.M., Rey-Mendez, M. and N. González-Henríquez. 2021. First report of the mussel *Mytella strigata* (Hanley, 1843) in the Venezuelan Caribbean from an invasion in a shrimp farm. Latin American Journal of Aquatic Research, 4 First report of 9(3): 531-537, 2021 *Mytella strigata* in the Venezuelan Caribbean 531. DOI: 10.3856/vol49-issue3-fulltext-2626
- Ma P-Z, Li H-M, Liu Y-M, Li C, Zhang Z, Wang H-Y (2022) First confirmed occurrence of the invasive mussel *Mytella strigata* (Hanley, 1843) in Guangdong and Hainan, China, and its rapid spread in Indo-West Pacific regions. BioInvasions Records 11(4): 947–963, <https://doi.org/10.3391/bir.2022.11.4.13>
- Mendoza, D.M., Aquino, M.G.B. and G.C. Mendoza. 2023. Shell Dimension-Weight Relationship and Condition Index of the Invasive Mussel *Mytella strigata* in Sasmuan, Pampanga, Philippines. / The Philippine Journal of Fisheries 30(1): 97-106. DOI: 10.31398/tpjf/30.1.2022-0022.
- Rice, M.A., Rawson, P.D., Salinas, A.D. AND W.R. Rosario. Identification and Salinity Tolerance of the Western Hemisphere Mussel *Mytella strigata* Charruana (DORBIGNY, 1842) in the Philippines. Journal of Shellfish Research, Vol. 35, No. 4, 865–873.
- Sanpanich K, Wells FE (2019). *Mytella strigata* (Hanley, 1843) emerging as an invasive marine threat in Southeast Asia. BioInvasions Records 8(2): 343–356, <https://doi.org/10.3391/bir.2019.8.2.16>
- Spinuzzi S, Schneider KR, Walters LJ, Yuan WS, Hoffman EA. Tracking the distribution of non-native marine invertebrates (*Mytella charruana*, *Perna viridis* and *Megabalanus coccopoma*) along the south-eastern USA. Marine Biodiversity Records. 2013;6:e55. doi:10.1017/S1755267213000316
- Teresa Stephanie Tay, Bin Qi Gan, Siew Chen Serina Lee, Chin Sing Lim, Koh Siang Tan & Serena Lay-Ming Teo (2018): Larval development of the invasive charru mussel, *Mytellastrigata* (Bivalvia: Mytilidae), Invertebrate Reproduction & Development, DOI: 10.1080/07924259.2018.1505671. To link to this article: <https://doi.org/10.1080/07924259.2018.1505671>.
- Yu Y, Gao Q, Liu M, Li J, Wang S, Zhang J (2023) Report on the invasive American brackish-water mussel *Mytella strigata* (Hanley, 1843) (Mollusca: Mytilidae) in Beibu Gulf. BioInvasions Records 12(1): 196–207, <https://doi.org/10.3391/bir.2023.12.1.16>
- Yuan, W., Walters, L.J., Schneider, K.R., and E.A. Hoffman. 2010. Exploring the Survival threshold: A Study of Salinity Tolerance of the Non-native Mussel *Mytella charruana*. Journal of Shellfish Research, Vol. 29, No. 2, 415–422, 2010.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F09
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	January 26, 2023
			Page	Page 6 of 6

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

Ianthe Marie P. Benliro – Faculty of the Fisheries Department under the College of Agriculture and Fisheries; conducted and co-authored studies on morphological identification of seaweeds, biodiversity of mangrove and mangrove-associated species as well as seagrass and seagrass associated organisms.

Gelene Roda I. Alvarez - Faculty of the Fisheries Department under the College of Agriculture and Fisheries; conducted studies in aquaculture specifically on blue swimming crabs, hatchery of milkfish and crabs, and fish nutrition.

Dr. Rey J. dela Calzada – Faculty of the Fisheries Department and dean of the College of Agriculture and Fisheries; published studies in peer-reviewed journals in fisheries particularly regarding studies on *Litopenaeus vannamei*, crabs and other aquaculture commodities.

Robert Christian F. Patani- Faculty of the Fisheries Department under the College of Agriculture and Fisheries; conducted studies on seaweed tissue culture, polyculture of aquatic organisms, nutrition, and biotechnology.

Alessandra Nicole B. Falalimpa - Faculty of the Fisheries Department under the College of Agriculture and Fisheries; conducted studies on post-harvest fisheries specifically fish protein hydrolysates, bio-fuels, product development of fisheries commodities.